MAT/CSC 210 | Chartier
Homework #4 | Spring 2026

Linear Programming Homework

Homework due Monday, March 2

Directions: This homework may be typed or handwritten.

1. Recall the problem of life vests and life boats from Chapter 1 of the class notes. For simplicity consider smaller boats with the parameters

$$\begin{aligned} C_1 &= 1 \quad \text{(capacity of each vest)} \\ C_2 &= 2 \quad \text{(capacity of each boat)} \\ C &= 10 \quad \text{(total capacity needed)} \\ V_1 &= 1 \quad \text{(volume of each vest)} \\ V_2 &= 5 \quad \text{(volume of each boat)} \\ V &= 15 \quad \text{(total volume available)} \end{aligned}$$

   (a) As an initial model, we will ignore the constraints that x_1 and x_2 should be integers. Also, we will ignore the constraint that x_2 should be an even number. Given these assumptions, set up the problem of maximizing the number of life boats that can be carried as a linear programming problem. Write the LP for this problem.

   (b) Use the "graphical method" to find the solution to the LP found above. Draw the feasible region in the x_1-x_2 plane and draw several contour lines of the objective function. (That is, draw several of the lines perpendicular to a line that is parallel to the gradient vector of the objective function.) Draw the contour line that intersects the corner(s) of the feasible set at which the optimum occurs. At what point(s) in the feasible region does the objective function attain its optimum?

   (c) Now, let us take another step in the modeling process and introduce the constraint that x_1 and x_2 must be integers. This means only a finite number of discrete points from the previous feasible set are still feasible (points for which x_1 and x_2 are both integers). For the feasible region found above, plot all feasible points that meet this integer constraint on x_1 and x_2. Solve this integer programming problem. Explicitly list the point(s) that attain the optimum.

   (d) Suppose the life boats are smaller and only require volume $V_2 = 2.5$. Repeat part (b) for this case.

   (e) Verify your solution in (b) using Excel. Submit a print-out of your Excel worksheet and the Answer Report that Excel produces.

2. A manufacturing firm has discontinued the production of a certain unprofitable product line. This act created considerable excess production capacity. Management is considering devoting this excess capacity to one or more of three products; call them products 1, 2, and 3. The available capacity on the machines that might limit output is summarized in the following table:

Machine type	Available time (in machine hours per week)
Mailing machine	500
Lathe	350
Grinder	150

The number of machine hours required for each unit of the respective products is:

Machine type	Product 1	Product 2	Product 3
Mailing machine	9	3	5
Lathe	5	4	0
Grinder	3	0	2

Note that the productivity coefficient in machine hours per unit. A coefficient of 0 indicates that the corresponding product does not require the particular machine type to be produced.

The sales department indicates that the sales potential for products 1 and 2 exceeds the maximum production rate and that the sales potential for product 3 is 20 units per week. The unit profit would be \$30, \$12, and \$15, respectively, on products 1, 2, and 3. The objective is to determine how much of each product the firm should produce to maximize profit.

(a) Write an LP formulation that would model this problem.

(b) Solve this LP using Excel, ignoring the fact that the number of each product should be an integer. With your write-up hand in an Excel Answer Report.

(c) Which of the constraints are "binding" at the optimimum? (E.g. which of the three machines are being used at maximum capacity and which are not?)

3. Mariko has many interests and wants to take a number of crafts and sports classes during the summer at the community center. In fact there are 7 different courses she would like to take. But each one requires 5 hours per week of commitment and costs \$15/week. She has a job which pays \$10/hour, and she must work enough to earn \$200/week plus the cost of any classes she takes. Her boss has told her that she can work any number of hours she chooses between 25 and 40 each week. She is willing to spend at most 55 hours a week between her job and any classes she takes. She wants to determine how many classes she can take.

   (a) Let x_1 represent the number of hours per week she works, and x_2 the number of hours per week she spends on classes. First, assume that both variables can take any value, which ignores the fact that x_2 must be a multiple of 5. Set up an LP that maximizes x_2 subject to some constraints. Write out all of the constraints mathematically and draw the feasible set in the x_1-x_2 plane. Indicate where the solution to the linear programming problem is.

   (b) The solution lies on the intersection of two constraints; list the two constraints. Further, find the solution by hand. That is, don't solve the problem using linear programming software, although you may do so, if you wish to check your answer.

   (c) Now, we introduce that added constraint that x_2 must be a multiple of 5. This will allow us to solve the actual problem to determine the number of classes that Mariko can take. Note that we have the additional constraint on x_2 but will allow x_1 to take any value. Is there a unique solution? If not, how many solutions exist and how might Mariko decide between the possible solutions?

       Note: This is called a *mixed integer programming problem* since it has a mixture of variable types, some of which must be integer, and the others continuous.